

RFQ - WSL-LANE-HARNESS-A Rev 4

RFQ — Custom Wire Harness Assembly (Lane Controller Field Harness)

Westside Lanes · Olympia, WA · Issued 2026-07-25 · Contact: Dylan DeYoung **Assembly: WSL-LANE-HARNESS-A · Rev 4**

Quantity: 1 first article → 2 pilot → balance to **34 total** (32 lanes + 2 spares). Option pricing requested at 48 and 64.

Rev 4 supersedes Rev 3 — quote against Rev 4 only. Neither Rev 2 nor Rev 3 was issued. **Rev 4 change: four new leads (W50–W53), two power pairs.** We are removing a legacy interface board from the machine and powering its sensors and a camera ourselves. **WARNING: The two pairs are electrically SEPARATE domains and must never be cross-connected** — see §6 and the wire-list notes. New colours **Violet/Grey** (isolated pair) and **Brown/Pink** (camera pair) are used for no other lead. Lead count 49 → 53.

Rev 3 change: the enclosure-to-machine run was measured at **6 ft (1829 mm)**. Rev 2 lead lengths were sized before that and were too short once the in-enclosure route is added. L1 2500→**3200**, L2 3000→**3700**, L3 4000→**4700**, L1-200 2300→**3000**. L0 (800) and the J14 jumper (120) are unchanged.

Rev 2 changes vs Rev 1 (all carried forward into Rev 3): (1) **All lead lengths increased** — Rev 1 budgeted the machine run but not the in-enclosure route (§3, §9). (2) **J14 ferrules corrected to bare/uninsulated** — an insulated 0.75 mm² collar fouls the adjacent pole at 3.5 mm pitch (§5). (3) **Terminal torque now specified** (§5). (4) **Strip length corrected 8 mm → 7 mm** (§5). (5) **Coding profiles added** to Tier 1 (§4). (6) **1840489 is customer-free-issued** (§4). (7) Two questions we previously asked you are now answered by us (§Questions).

0. Read this first — what kind of customer we are

We are a **small business (a bowling center), not an OEM**. We do not require PPAP, a formal FAI package, ISO quality-system flow-down, a supplier audit, or source inspection. **Your standard Certificate of Conformance is acceptable.** Please do not price a documentation program into this quote — if you see a line item that exists only to satisfy an aerospace or automotive customer, leave it out and tell us you did.

What we *do* care about, in priority order: **correct and unambiguous labeling**, 100% electrical test, consistency unit-to-unit, and on-time delivery.

1. What this assembly is

A field wiring harness for a replacement control board on bowling pinsetter machines. One harness per lane. It connects a controller PCB (mounted in a wall enclosure) to sensors, switches, and relay-coil circuits on the machine. The enclosure-to-machine run is **measured at 6 ft (1.83 m)**; leads are built longer than that to cover the in-enclosure route plus service slack.

Electrically trivial — nothing above 33 V, nothing above ~1 A. This is a **labeling- and termination-intensive** assembly, not a high-voltage or high-current one. Cost lives in the ~53 leads × 2 ends of printed markers, not in the wire.

Environment: indoor, unconditioned equipment area behind bowling pinsetters. Ambient to ~35 °C. Fine wood/pin dust and light lane-oil mist. Vibration from adjacent machinery. No flexing in service once installed — this is a static installation.

2. Scope boundary (important)

Quote the **controller-side harness only**, as specified here:

- **Controller end (one end of every lead): FULLY TERMINATED** — insulated ferrule, landed and torqued into its Phoenix plug, per §4/§5.
- **Machine end (the other end): PREPARED BUT UNTERMINATED** — cut to length, labeled, and left with **10 mm of insulation stripped, no ferrule** unless noted otherwise in the wire list. We terminate the machine end ourselves on site (the machine-side interface is a legacy 1982 connector still being characterized).

Do **not** quote any machine-side connector. If you have an opinion on whether the stripped ends should instead ship unstripped, say so.

3. Wire list

Attached as a separate CSV (`WSL-LANE-HARNESS-A_wirelist_rev4.csv`), one row per lead, with columns: `Wire ID` · `AWG` · `UL style` · `Color` · `End-A connector ref + position` · `End-A termination` · `End-B termination` · **`**Length Class**`** · `Finished length` · `Label text (identical both ends)` · `Twisted-pair partner` · `Notes`.

Rev 2 added a Length Class column; Rev 3 revises the class values. Every lead belongs to one of four classes (**L0 / L1 / L2 / L3**). If we revise lengths again, only the class values change — the per-lead assignments stay put. Please structure your quote so a class-length change is a price delta, not a re-quote.

Summary for scoping:

Group	Connector (controller end)	Leads	AWG	Length class	Length
Fast field inputs — cams	J3 — Phoenix MC 1,5/10-ST-3,5	5 (positions 1–5; all capped)	22	L3	4.7 m
Fast field inputs — DIELL	J3 (positions 7–10; 6 empty)	4	22	L2	3.7 m
Slow inputs A	J4 — MC 1,5/14-ST-3,5	13 (positions 1–11, 13, 14; 12 empty)	22	L1	3.2 m
Slow inputs B	J5 — MC 1,5/12-ST-3,5	4 (positions 1, 2, 3, 12; 4–11 empty)	22	L1	3.2 m (FOUL lead 3.0 m)
Lamp outputs	J13 — MC 1,5/ 6-ST-3,5	6	22	L3	4.7 m (see §9)
Safety loop	J14 — MC 1,5/ 4-ST-3,5	2 + 1 internal jumper	18	L1	3.2 m (jumper 120 mm)
Power in	none — ferruled loose leads	3	18	L0	0.8 m
Machine outputs	none — ferruled loose leads	12	18	L1	3.2 m
Sensor power (isolated) WARNING:	none — ferruled loose leads	2	18	L2	3.7 m
Camera power WARNING:	none — ferruled loose leads	2	18	L3	4.7 m
Total		~53 leads			

Length class definitions: **L0** = stays inside the enclosure · **L1** = enclosure to the machine connector · **L2** = enclosure to the photoelectric sensors · **L3** = enclosure to the overhead lamp unit / far mechanism.

4. BOM

Tier 1 — NO SUBSTITUTIONS (form/fit/function critical)

Item	Manufacturer	Part number	Qty/assy	Source
Plug, 10-pos, 3.5 mm	Phoenix Contact	1840447 (MC 1,5/10-ST-3,5)	1	you
Plug, 14-pos, 3.5 mm	Phoenix Contact	1840489 (MC 1,5/14-ST-3,5)	1	WARNING: CUSTOMER FREE- ISSUE
Plug, 12-pos, 3.5 mm	Phoenix Contact	1840463 (MC 1,5/12-ST-3,5)	1	you
Plug, 6-pos, 3.5 mm	Phoenix Contact	1840405 (MC 1,5/ 6-ST-3,5)	1	you
Plug, 4-pos, 3.5 mm	Phoenix Contact	1840382 (MC 1,5/ 4-ST-3,5)	1	you
Coding profile	Phoenix Contact	1734634 (CP-MSTB)	2	you

****WARNING:** 1840489 is FREE-ISSUED — do not source it.** It is discontinued and the authorized distribution channel is dry. We are making a lifetime buy and will ship it to you with the PO. Tell us your required buffer quantity. **Pre-approved fallback** if our stock runs short: **Phoenix FMC 1,5/14-ST-3,5** (push-in rather than screw; mates the identical MCV 1,5-G header, no other change). Use it only if we authorize in writing.

Coding profiles (new in Rev 2) — install 2 per assembly: J3 at pole 1 and J13 at pole 1. Profiles fit the **PLUG**, never a header. This is not cosmetic: the board has two pairs of *identical* connectors (J3/J15 both 10-pos, J13/J16 both 6-pos). An uncoded plug mates either one. A J13 lamp plug pushed into J16 lands LEDs across 5 V and wedges the I²C bus. Also apply the **WHITE identification band** per the marking scheme in the attached harness BOM. **Prove the coding operation on a sacrificial plug before coding production parts** — profile installation is not reversible.

Tier 2 — functional equivalents acceptable (tell us what you'd use)

Hook-up wire (UL1007 or UL1015 class, 300 V min for 22 AWG, 600 V for 18 AWG), insulated ferrules, printed heat-shrink marker stock, adhesive-lined heat-shrink end caps, lacing/ties, bags.

Wire colors are functionally meaningful — do not substitute colors. See §6.

5. Ferrule, strip, and torque specification

Strip length is 7 mm on every landed end. (Rev 1 said an 8 mm ferrule barrel — that was wrong; 7 mm is the Phoenix MC 1,5 figure.)

Leads	Lands in	Pitch	Ferrule
22 AWG → J3, J4, J5, J13	MC 1,5-ST-3,5 plugs	3.5 mm	0.34 mm² INSULATED , 7 mm
18 AWG → J14 only (JMP1, W33, W34)	MC 1,5/4-ST-3,5	3.5 mm	WARNING: BARE STRANDED, or 0.75–1.0 mm² UNINSULATED , 7 mm
18 AWG → loose leads for J2 / J6–J11	MKDS 1,5 blocks	5.08 mm	0.75 mm² INSULATED , 7 mm

****WARNING:** Why J14 is different — this is the one place Rev 1 was wrong.** The Phoenix MC 1,5 series is rated 0.14–1.5 mm² **bare or with an uninsulated ferrule, but only 0.5 mm² maximum with an insulated ferrule**. A 0.75 mm² insulated collar is roughly 4.0 mm across, against a 3.5 mm pole pitch — and all four J14 poles are populated, so adjacent collars physically interfere. **Do not fit insulated ferrules on any J14 lead**. The 0.75 mm² insulated ferrules on the loose J2/J6–J11 leads are correct and must stay: those blocks are 5.08 mm pitch, rated to 1.5 mm². Please do not “standardize” the three J14 leads to match them.

Terminal screw torque: 0.22–0.25 N·m (M2 screw). This is roughly **36 landed screws per assembly**. Do not use ~0.5 N·m — it is more than 2× the rating and will strip the screw or crack the plug body. Use a calibrated driver.

- Phoenix AI-series or Weidmüller H-series preferred; equivalents acceptable.
- Crimp to the ferrule manufacturer’s die spec; **every ferruled joint must pass a pull test**.

6. Wire colors (functional, not cosmetic)

Color	Meaning
White	landed sense signal
Orange	DO-NOT-LAND lead — capped at the machine end, deliberately unconnected until a future commissioning step. Orange must not appear on any other lead.
Green	field ground / chassis bond
Black	ground / return
Blue, Yellow (22 AWG)	photoelectric sensor signals L and R
Red (18 AWG)	relay normally-open output, and +5 V power
Black (18 AWG)	relay common, and power return
Yellow (18 AWG)	safety-loop circuit
Violet	isolated +12 V — sensor supply. WARNING: separate electrical domain
Grey	isolated 0 V — sensor return. WARNING: separate electrical domain
Brown	camera +12 V (non-isolated)
Pink	camera 0 V (non-isolated)

WARNING: The Violet/Grey pair and the Brown/Pink pair are two SEPARATE electrical domains and must never be joined, spliced, or share a ferrule. They are deliberately given four unique colours, used on no other lead, so the distinction survives installation. Please keep them in **separate bundles** and do not lace them together.

7. Labeling — the cost driver, specified precisely

- **Printed heat-shrink sleeve markers** (Brady PermaSleeve, Phoenix THERMOMARK, or equivalent), white sleeve, black print, permanent. Not wrap-around adhesive labels; not hand-marked.
- **BOTH ends of every lead**, ~53 leads = **~106 markers per assembly**.
- Exact text is in the wire-list CSV, column `Label text`. Text is identical at both ends of a given lead. **Labels are deliberately short — longest string is 22 characters** (`STOP/CIS - DO NOT LAND`). Signal names only; no machine-specific wiring references appear on the wire, because those vary per lane and live on a per-lane card instead. If any string is still too long for your marker stock, tell us your maximum — do not truncate on your own initiative.
- Position: within 40 mm of each termination, oriented so text reads with the lead running left-to-right.
- Additionally: each of the five Phoenix plug bodies carries its own printed label (text in the CSV, `Plug body label` sheet).

8. Special construction features

1. **Twisted pairs (2 per assembly)**. Blue+Black and Yellow+Black on J3, ~1 twist per 30–40 mm, starting 50 mm from the plug and stopping **100 mm short of the machine ends** so those ends remain independently routable. Pair partners are named in the wire list.
2. **Capped leads (11 per assembly)**. All orange leads plus the two 18 AWG yellow safety leads get their **machine end sealed with adhesive-lined heat shrink** — no ferrule, no exposed conductor. They still carry their printed label. These are deliberately dead-ended.
3. **Empty connector positions**. J3-6, J4-12, and J5 positions 4–11 are **intentionally empty**. Do not populate them. The wire list is authoritative; if a position isn't listed, it stays empty.
4. **J14 internal jumper**. A ~120 mm 18 AWG yellow U-jumper is fitted **inside the J14 plug between positions 1 and 2**, ferruled both ends, with a printed flag label on the loop (text in the CSV). This is a deliberate engineered link, not an error. Positions 3 and 4 carry the two capped leads. **A finished J14 must read SHORT across 1–2 and OPEN across 3–4**.
5. **Loose ferruled leads**. The 3 power leads and 12 machine-output leads have **no connector at the controller end** — they ship ferruled and labeled, loose, to be landed in screw terminals at install. Bundle and tie them per group.

9. Lengths — build long, we trim on site

Rev 3 lengths are measured-plus-margin. The enclosure-to-machine run is **6 ft (1829 mm)**, measured. Each L1 lead must also cross the in-enclosure route (~750–950 mm on the far board) plus ~300 mm service slack and drip loop, which is how 3200 mm is derived. Per-lane siting is not yet surveyed at every lane pair, so all four classes are sized so that **no lead can come up short**. We will trim on site. Build to the wire list; do not optimize lengths down.

****WARNING:** Marker placement requirement (new in Rev 2, and it affects your process).** Because we trim the machine end in the field, a marker printed at the very end of the lead gets cut off. On every **L1 / L2 / L3** lead, either: **(a)** position the machine-end marker **150 mm back from the cut end**, or **(b)** ship the machine-end markers loose, unshrunk, in a labeled bag per assembly. **State which you will do**. The controller-end marker is unaffected. (Correct labeling is priority #1 in §0 — a trimmed-off marker defeats the whole spec.)

Please quote: **(a)** at the Rev 3 lengths as written, and **(b)** the per-assembly cost delta per **±0.5 m on class L1** (30 leads), so we can price the classes down once the site survey is complete without re-quoting.

10. Test and workmanship

- **Workmanship: IPC/WHMA-A-620, Rev E, Class 2.** Class 2 is correct for this application — please do not build or price to Class 3.
- **100% electrical test on every assembly:** point-to-point continuity on all nets, **plus isolation/shorts testing between all nets.** Isolation matters more than usual here — an adjacent-position short inside a Phoenix plug is our most likely failure mode.
- **Explicit exception:** the J14 1–2 jumper is an intentional short and must not be flagged.
- **No hipot required** (max circuit voltage 33 V).
- Ship a test record or pass/fail log per serialized assembly.

11. Identification and traceability

- Each assembly carries a printed label with **assembly PN + revision + serial number** on the main bundle near the J4 plug.
- Serial-level traceability requested (simple sequential numbering is fine — we need to tie a harness to a lane, not to a wire lot).

12. Quantity, ramp, and spares

Stage	Qty	Note
First article	1	Ship for our approval before proceeding. We will respond within 5 business days.
Pilot	2	After FA approval
Balance	31	32 lanes + 2 spares total
Total program	34	Please also quote 48 and 64 so we can see the price break

Additionally quote as separate line items: **10% spare loose leads** (assorted, pre-labeled) and **3 spare sets of the five Phoenix plugs**, unpopulated.

13. Packaging and kitting

Individually bagged and labeled per assembly, with the serial number visible on the outside of the bag. This is worth paying for on a 32-lane install — it is what prevents a wrong-harness-on-wrong-lane afternoon. Coil to ~250 mm diameter, tie in at least 3 places, one carton per 6–8 assemblies with a packing list.

14. Commercial

- **Target:** first article within 3 weeks of PO; balance within 8 weeks of FA approval.
- Ship to Olympia, WA 98502. Freight prepaid and added is fine.
- Quote NRE separately from unit price, and tell us what the NRE buys (fixture, program, setup) and whether it recurs on a reorder.

- State your acceptable over/under shipment tolerance.
- **Drawing revision control stays with us.** We will issue revised wire lists as `Rev 2`, `Rev 3`, etc.; please quote against the revision stated on the PO.

15. Attachments provided with this RFQ

1. `WSL-LANE-HARNESS-A_wirelist_rev4.csv` — the 53-lead from-to wire list with label text and length classes. **This is the controlling document. Where it and this RFQ disagree, the CSV wins.**
2. `phase8_revD_harness_bom.csv` — plug + coding-profile part identities and the band-marking scheme
3. Phoenix Contact datasheets for the five Tier-1 plugs and the CP-MSTB coding profile

Not attached, and deliberately so: Rev 1 referenced a dimensioned 2D layout PDF and a golden-sample photo set. **Neither exists** — the prototype unit has not been built. Both references are withdrawn rather than left dangling. If a flat layout drawing would materially reduce your quote, say so and we will produce one.

Plug-body labels: the J14 plug body carries `J14 SAFE LOOP - SHORT 1-2` (text is in the wire list). The other four plug bodies need identification text we have **not** yet specified — quote plug-body labeling as a **separate line item** and we will supply the text on PO.

Questions we specifically want answered in your quote

1. Your maximum legible characters per heat-shrink marker (§7).
2. The §9 length delta (per ± 0.5 m on class L1), and **which marker-placement option you will use** — set back 150 mm, or shipped loose and un-shrunk.
3. Your required **buffer quantity** for the free-issued 1840489 (§4).
4. Confirmation that you can install **CP-MSTB coding profiles** (§4), and whether you would rather we ship the plugs pre-coded.
5. Anything in this package that is over-specified and costing us money for no benefit. We would rather hear that than pay for it.

Two Rev-1 questions are now answered by us — no response needed: **Ferrules (old Q1):** resolved in §5. J14 takes bare or uninsulated; everything else is as tabulated. Do not substitute. **Machine-end preparation (old Q4):** build them **stripped 10 mm as specified**. The machine end is crimped into a contact on our side, so stripped is correct.